ADVANCED CYBER LAW, FORENSICS & CRIMINAL JURISPRUDENCE

MODULE 1: Indian Evidence Act, 1872 & Bharatiya Sakshya Adhiniyam, 2023 (BSA)

1.1 The Epistemological Nature of Electronic Evidence (Computer Science Perspective)

Digital evidence comprises discrete binary signals represented on electromagnetic, optical, or solid-state silicon media. Unlike analog evidence (ballistics, fingerprints, handwritten documents), electronic records possess distinct forensic properties that complicate legal admissibility:

- **Zero Marginal Alteration Cost:** A single flipped bit can invalidate an encrypted signature or modify an executable payload without leaving physical traces.
- **Volatility & Transience:** Data residing in cache, register banks, and RAM diminishes instantaneously upon power loss (RFC 3227 Volatility hierarchy).
- **Separation of Container and Content:** The file system (NTFS, EXT4, APFS) decouples physical block allocation from logical data representation, requiring specialised bit-stream acquisition rather than simple file copying.

1.2 The Statutory Bridge: Section 65A & 65B (IEA) vs. Section 61 & 63 (BSA 2023)

Under the Indian Evidence Act (IEA 1872), the general provisions for proving documents (Section 63 & 65) are displaced by Section 65A and 65B for electronic evidence. This statutory architecture is codified in Section 61 and Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (BSA).

- **The Legal Fiction (Sec. 65B(1) / BSA Sec. 63(1)):** A computer output—whether a physical printout, a write-blocked bit-stream clone, an optical disc, or a pen-drive dump—is legally deemed to be a 'document' and is admissible as secondary evidence without demanding production of the original multi-rack physical server.
- **The Four Technical Conditions (Sec. 65B(2) / BSA Sec. 63(2)):**
  - Lawful Custody & Regular Production: The system was actively and lawfully used by the custodian during the period in question.
  - Regular Ingestion in Ordinary Course: Data was fed systematically into the system as part of ordinary business or automated logging routines.
  - System Operational Integrity: The hardware/software was operating properly throughout the period, or any transient outage/malfunction did not compromise the bit-level integrity of the stored record.
  - Derivative Authenticity: The output accurately reflects or derives directly from the ingested data.

1.3 The Mandatory Certificate: IEA Sec. 65B(4) vs. BSA 2023 Schedule (Two-Part Certificate)

Under Section 65B(4) IEA, secondary electronic evidence was rejected if not accompanied by an authentic certificate signed by an authorised system manager. However, lack of technical standardisation led to widespread certificate forgery and administrative ambiguity.

BSA 2023 Technological Modernisation: The BSA 2023 reform formally structures this requirement into a statutory Schedule containing two distinct certifications:

- **Part A (Administrative / Custodial Certification):** Executed by the person managing the computer device or official responsible for the automated logging process, verifying custody, routine operation, and system health.
- **Part B (Forensic / Technical Expert Certification):** Executed by a digital forensic examiner or technical specialist. Crucially mandates the record's Cryptographic Hash Value (e.g., SHA-256 / SHA-3), forensic tool chain verification (EnCase, FTK, Autopsy, dd), write-blocking logs, and bit-stream verification to establish mathematically that the evidence has suffered no post-seizure tampering.

1.4 Landmark Judicial Trajectory (Evolution of Digital Evidence Law)

- **State (NCT of Delhi) v. Navjot Sandhu (2005) 11 SCC 600:** The Supreme Court held that secondary electronic evidence (Call Detail Records) could be proven via oral evidence and general Section 63/65 provisions without a 65B certificate. This opened severe security loopholes in evidence handling.
- **Anvar P.V. v. P.K. Basheer (2014) 10 SCC 473:** A 3-judge bench definitively overruled Navjot Sandhu, holding that Sections 65A and 65B form a complete, specialized code. Electronic records cannot bypass Section 65B(4). Without the certificate, secondary electronic records are strictly inadmissible.
- **Shafhi Mohammad v. State of H.P. (2018) 2 SCC 801:** Introduced temporary confusion by carving an exception for parties not in physical control of the device.
- **Arjun Panditrao Khotkar v. Kailash Kushanrao Gorantyal (2020) 7 SCC 1:** The definitive three-judge bench verdict settled the law: (1) The Section 65B(4) certificate is an absolute condition precedent for secondary electronic records; (2) If the original device (primary physical hardware, e.g., smartphone, DVR, laptop) is produced in court, no certificate is needed; (3) If an ISP or third-party custodian refuses to supply the certificate, the trial court has a mandatory duty to issue summons under Section 91 CrPC.

MODULE 2: Indian Penal Code, 1860 & Bharatiya Nyaya Sanhita, 2023 (BNS)

2.1 Dual-Charging Doctrine: Technical Offenses vs. Substantive Mens Rea

Students must understand that while the Information Technology Act, 2000 penalizes cyber-dependent acts (unauthorized access under Sec. 43/66, cracking, tampering with source code under Sec. 65), contemporary cybercrimes are overwhelmingly cyber-enabled economic, social, or organized crimes. Prosecutors routinely invoke both the IT Act and IPC (now BNS) to address both the computational mechanism and the substantive criminal intent (fraudulent enrichment, extortion, criminal conspiracy).

2.2 Mapping Matrix: IPC vs. BNS (2023) in Cyber Scenarios

Offence Class	IPC (1860)	BNS (2023)	Computer Science & Forensic Manifestation

Cheating & Fraud	Sec. 420 (Cheating & dishonestly inducing delivery)	Sec. 318(4)	Spear-phishing, fake payment gateways, malicious APK banking trojans, rogue APK overlays, SIM swap frauds, algorithmic pig-butchering scams.
Personation / Identity Theft	Sec. 416 & 419 (Cheating by personation)	Sec. 319	Deepfake audio synthesis, AI voice cloning, caller ID spoofing, business email compromise (BEC), unauthorized use of digital identities.
Digital Forgery	Sec. 463, 465, 468, 471 (Forged electronic record)	Sec. 336, 338, 340	Forging X.509 PKI certificates, tampering with SMTP mail headers, synthetic biometric generation, fabrication of PDF digital signatures, altering ledger entries.
Cyber Extortion / Ransomware	Sec. 383, 384 (Extortion)	Sec. 308	Ransomware operations (AES-256 + RSA-4096 hybrid encryption), double/triple extortion schemes, AI deepfake morphing / sextortion rackets.
Cyberstalking & Surveillance	Sec. 354D (Cyberstalking)	Sec. 78	Commercial spyware injection (zero-click exploits, Pegasus-style toolchains), unauthorized Bluetooth/AirTag geolocation tracking, persistent automated OSINT stalking.
Organized Cyber Syndicates	Sec. 120B (Conspiracy) + General IPC	Sec. 111 (New) (Organized Crime - Cyber Inclusion)	Distributed darknet syndicates, Ransomware-as-a-Service (RaaS) affiliate networks, multi-layered mule account laundering chains, credential-stuffing botnets.

MODULE 3: Code of Criminal Procedure, 1973 & BNSS, 2023 (Procedural & Forensic Investigation)

3.1 Topological vs. Geographic Jurisdiction (Sec. 177–189 CrPC vs. BNSS)

A fundamental challenge in cyber investigation is the borderless nature of packet-switched networks:

- **Multi-Jurisdictional Locus (Sec. 178 & 179 CrPC / BNSS Sec. 198 & 199):** Where an offense comprises acts committed across different server nodes, cloud regions, and client endpoints, jurisdiction lies in any court where any component act occurred, or where the victim experienced the direct computational or financial consequence.

- **Extraterritorial Jurisdiction (Sec. 188 CrPC / Sec. 75 IT Act):** Applies to offenses committed outside India by Indian citizens, or by foreign nationals targeting computer systems, networks, or data located within India. Investigation requires Central Government sanction under Section 188.

3.2 Search, Seizure & Preservation of Digital Media (CrPC vs. BNSS 2023)

- **Section 91 CrPC (BNSS Sec. 94):** Production notices to Intermediaries (AWS, Cloudflare, Meta, Google) and Telecom Providers (TSPs) demanding IPDR (IP Detail Records), NAT translation logs, DHCP logs, and radius authentication logs. Time is critical due to statutory data retention caps.
- **Section 102 CrPC (BNSS Sec. 107):** Seizure of computational hardware (NVMe drives, routers, HSM tokens) and freezing of illicit financial assets (UPI handles, crypto wallet addresses, central exchange accounts).
- **Mandatory Audio-Visual Recording (BNSS Sec. 105):** A major reform under BNSS 2023: Search and seizure procedures, including the physical unmounting, packaging, and tagging of digital hardware, must be recorded via electronic/audio-visual devices. This protects against allegations of forensic contamination or planting of malicious artifacts.

3.3 Forensic Standards for Digital Evidence Acquisition

For an M.Tech cyber forensics professional acting as an expert witness (Sec. 45 IEA / Sec. 39 BSA):

- **Order of Volatility (RFC 3227):** CPU Registers/Cache → Live RAM (DRAM) → Network State / ARP Tables / Netstat → Active Process Tables → Temporary Swap Files → Secondary Storage (SSD/HDD) → Remote Archival Logs.
- **Hardware Write-Blockers:** Ensures bidirectional communication commands (write, delete, format) are intercepted at the hardware controller level during disk imaging.
- **Forensic Image Verification:** Bit-stream raw copies (.dd, .raw) or forensic containers (.E01, .AFF4) must be validated with matching SHA-256 / SHA-3 cryptographic hashes calculated before and after acquisition.

3.4 Statutory Conflict: Special Law (IT Act) vs. General Law (IPC) Precedence

In Sharat Babu Digumarti v. Govt. of NCT of Delhi (2017) 2 SCC 18, the Supreme Court invoked Section 81 of the IT Act (non-obstante clause) and the principle of Lex Specialis Derogat Legi Generali. When an electronic transmission constitutes an offense explicitly covered under Section 67/67A/67B of the IT Act, the prosecution cannot invoke Section 292 of the IPC to circumvent the IT Act's specific statutory framework. However, where independent elements of cheating (Sec. 420 IPC) or extortion (Sec. 384 IPC) exist alongside cyber methods, composite charging remains valid.

MODULE 4: Pedagogical Case Scenarios & Examination Questions for M.Tech Students

Scenario A: Ephemeral Cloud Container Breach & Section 63 BSA Hash Admissibility

A microservices-based banking backend deployed on AWS Kubernetes (EKS) suffers a remote code execution (RCE) breach via an unpatched dependency. An attacker extracts PII data from an RDS database. The DevOps engineer terminates the affected container pods, spins up new replicas, and

downloads VPC flow logs and CloudTrail JSON files to an external laptop, printing them for police submission.

- **Problem for Students:** Evaluate the admissibility of the printed CloudTrail logs under Section 65B IEA / Section 63 BSA. Who qualifies to sign Part A and Part B of the BSA Schedule? Discuss the evidentiary implications of failing to capture a live RAM dump prior to pod termination.

Scenario B: Cross-Border AI Deepfake Extortion & Jurisdiction

An offshore attacker uses an open-source generative diffusion model hosted on a bulletproof VPS in Switzerland to create a deepfake video of a corporate executive in Indore, demanding 5 Monero (XMR). Communications occur via encrypted Matrix protocol.

- **Problem for Students:** Frame charges under the IT Act (Sec. 66D, 66E, 67A), BNS (Sec. 308, 319, 336), and determine court jurisdiction under Section 179 CrPC. Detail the Letters Rogatory (LR) and Mutual Legal Assistance Treaty (MLAT) procedural requirements for cross-border server seizure.

Here is a detailed breakdown and pedagogical explanation of both Case Studies in clear English, structured specifically for your M.Tech students.

Case Study 1: Ephemeral Cloud Container Breach & Section 63 BSA Hash Admissibility

1. The Scenario
Context: An enterprise runs a microservices banking backend deployed on **AWS Elastic Kubernetes Service (EKS)**.

The Attack: A threat actor discovers an unpatched third-party dependency and exploits a **Remote Code Execution (RCE)** vulnerability, enabling unauthorized exfiltration of Personally Identifiable Information (PII) from an Amazon RDS database.

The Response: Upon detecting abnormal traffic, the DevOps team immediately terminates the affected container pods, deploys clean replicas, downloads VPC Flow Logs and AWS CloudTrail JSON logs to a personal workstation, prints them out, and hands the paper printouts to law enforcement.

2. Core Legal & Forensic Issues

A. Primary vs. Secondary Electronic Evidence

- **Legal Principle:** Under Section 65B of the Indian Evidence Act (IEA) and Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (BSA), computer outputs (printouts, copies saved onto flash drives or external laptops) are **secondary evidence**.

- **Court Implication:** The printout cannot simply be submitted on its own. It is legally admissible **only if accompanied by a statutory certificate** verifying system integrity and lawful custody.

B. The BSA 2023 Two-Part Certificate Architecture

Under Section 63 of the BSA (read with its Schedule), the single, ambiguous certificate under old Section 65B(4) IEA has been replaced by two distinct components:

1. **Part A (Administrative / Custodial Certification):**
    - *Who signs:* The AWS Cloud Administrator or Enterprise System Manager who controls the cloud infrastructure and automated logging systems.
    - *What they attest:* That AWS CloudTrail and VPC Flow Logging operated routinely in the ordinary course of business, and there were no server outages affecting log integrity during that timeframe.
2. **Part B (Forensic / Technical Certification):**

    Who signs: A certified Digital Forensic Examiner or Technical Expert.

    - *What they attest:* The technical verification of how the logs were extracted, tool validation (e.g., hash generation via `sha256sum`), write-blocker logs if physical drives were used, and bit-level integrity confirmation.

C. The Digital Forensics Pitfall (Loss of Volatile Evidence)

- **Violation of RFC 3227 (Order of Volatility):**

  Terminating the Kubernetes container destroyed the **volatile runtime state** (RAM, dynamic process table, active sockets, decrypted memory payloads).

- **Forensic Defense Challenge in Court:**

  The defense attorney will argue:

    - Since the container memory was not captured prior to termination, the prosecution cannot establish the exact process lineage of the exploit.
    - Downloading JSON logs to an unmanaged personal laptop breaks the **Chain of Custody (CoC)**; the logs could have been modified before printing unless a cryptographically verifiable SHA-256 hash was generated at the exact moment of export from AWS S3/CloudWatch.

Case Study 2: Cross-Border AI Deepfake Extortion & Jurisdictional Multiplicity

1. The Scenario
Context: An offshore cyber criminal rents a bulletproof Virtual Private Server (VPS) located in Germany/Switzerland.

The Attack: Using open-source Generative Diffusion and voice-cloning models, the attacker creates a non-consensual synthetic nude video (deepfake) of a corporate leader based in Indore, India.

The Extortion: The attacker contacts the victim via an end-to-end encrypted Telegram/Matrix chat, threatening to leak the video across social platforms unless **5 Monero (XMR)** or **2 ETH** is transferred to a private cryptocurrency wallet.

2. Core Legal & Procedural Mapping

A. Statutory Charge Framing (Dual-Charging Matrix)

Governing Law	Relevant Section	Legal Justification
IT Act, 2000	**Section 66D**	Cheating by personation using a computer resource.
IT Act, 2000	**Section 66E**	Violation of privacy (capturing/publishing private images without consent).
IT Act, 2000	**Section 67A**	Publishing or transmitting sexually explicit material in electronic form.
BNS, 2023 *(formerly IPC)*	**Section 308** *(IPC 383/384)*	**Extortion:** Putting a person in fear of injury/reputational injury to induce property transfer.
BNS, 2023	**Section 319** *(IPC 416/419)*	Cheating by personation through synthetic digital media.
BNS, 2023	**Section 336 / 340** *(IPC 468/471)*	Digital forgery and using forged electronic records as genuine.
BNS, 2023	**Section 78** *(IPC 354D)*	Cyberstalking and non-consensual surveillance/harassment.

B. Territorial Jurisdiction: Topological vs. Geographical Dissonance

- **The Conflict:** The attacker is outside India; the server is in Europe; the victim and financial damage are in Indore, Madhya Pradesh.
- **Section 179 CrPC / BNSS Section 199:**

  When an act is an offense by reason of anything done and of a consequence that has ensued, jurisdiction lies with the court within whose local jurisdiction **the act was done or the consequence ensued**. Because the extortion demand was received and caused distress in Indore, the **Cyber Crime Police Station / Court in Indore possesses lawful territorial jurisdiction**.

- **Section 75 of the IT Act, 2000 & Section 188 CrPC:**

  Extends extraterritorial reach to offences committed outside India by any person, provided the computer system or network targeted is located in India. (Proceeding with trial against a foreign national requires Central Government sanction under Section 188 CrPC).

C. Cross-Border Evidence Acquisition & Extradition

- **Mutual Legal Assistance Treaty (MLAT) / Letters Rogatory (LR):**

  Since the hosting VPS is in Germany/Switzerland, Indian law enforcement cannot unilaterally hack or seize foreign physical servers. The investigating officer (IO) must route an evidence-preservation and search warrant request through the **Ministry of Home Affairs (MHA)** under the bilateral MLAT framework.

- **Blockchain Forensics:**

  Traceability analysis on the public ledger (for ETH) and monitoring off-ramps (centralized crypto exchanges / KYC endpoints) to link the pseudonymous wallet address to real-world identities.